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Roll No

EX-8002 (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Power Quality Problems and Mitigation Techniques

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss the general classes of power quality problems. What are the various solutions for Power quality improvement at end user level.
b) Define the following terms
 - i) Sag and Swell
 - ii) Voltage surge
 - iii) Flicker
 - iv) Distortion factor
 - v) Fast Fourier transforms
2. What are the series and shunt compensators compare their role for power quality improvement?
3. What do you understand by unified power quality compensator, discuss their classifications, working and operations?
4. Explain the following:
 - i) DSTATCOM
 - ii) Power quality evaluation procedure
5. a) Describe the production of harmonics from commercial loads . How it is different from industrial loads.
b) How harmonic sources can be modeled? Explain.
6. a) What is the need for estimating sag performance? Discuss in details about sag performance evaluation indices.
b) What are the various causes of voltage and current harmonics? Explain individual harmonic distortion and total harmonic distortion.
7. a) Describe the concept and working of active filters.
b) How can utilities deal with problems related to capacitor switching transients? Explain.
8. Explain the following:
 - i) Shunt active and Passive filters
 - ii) Active series compensators
 - iii) Harmonic reduction and standards
